2024

Airtel Digital Ltd Lowers Compute Spend for Music Streaming Service by 70% Using AWS

Airtel Digital Ltd transforms the cost-effectiveness of delivering Wynk Music by adopting services such as Amazon EC2 Spot Instances and moving to AWS Graviton Processors.

Overview | Opportunity | Solution | Outcome | AWS Services Used

20%

reduction in costs-per-user

70%

dec

60%

reduction in analytics platform expense

47%

reduction in costs of delivering APIs

66%

consolidation of clusters

Overview

Airtel Digital Ltd. (ADL) operates Wynk Music, a music streaming service in India. To optimize costs and free up resources for innovation, Wynk Music moved to microservices with Amazon Elastic Kubernetes Service (Amazon EKS) and switched to Amazon Elastic Compute Cloud (Amazon EC2) instances powered by AWS Graviton Processors and Amazon EC2 Spot Instances.

As a result of the optimization, Wynk Music has reduced costs for computing by 70 percent and the expense of supporting application programming interfaces by 47 percent. In addition, the company has lowered the cost-per-user by 20 percent, freeing up resources to re-invest in service development.

Opportunity | Driving Innovation at Scale to Improve Operations

Airtel Digital Ltd. (ADL) is the digital subsidiary of Bharti Airtel, a multinational telecommunications company based in India. The market for ADL services in India continues to grow, with the value of the country economy expected to be worth $1 trillion by 2030. As a result, ADL is rapidly developing new consumers and businesses and enhancing existing ones.

To drive innovation at scale and improve performance, ADL uses Amazon Web Services (AWS) to run the applications behind its offerings, ranging from a data-driven advertising platform and an omnichannel contact center to a data analytics infrastructure for customer insights.

The business continues to work closely with AWS to deploy new software and services into its cloud environments to optimize costs and improve the customer experience. This collaboration spans all ADL digital services, including Wynk Music, a music streaming service. Hitesh Bhatia, associate vice president of DevOps at Wynk Music says, “Our relationship works so well because AWS understands our goal of improving operations and freeing up resources for development.”

Solution | Enhancing Performance while Lowering Costs Using AWS

In 2021, Wynk Music rearchitected its applications to run in microservices-based containerized environments, with Amazon Elastic Kubernetes Service (Amazon EKS) to scale scheduled Amazon Elastic Compute Cloud (Amazon EC2) instances. It then moved from Amazon EC2 On-Demand instances to Amazon EC2 Spot Instances, which resulted in a reduction of about 10–15 percent on computing costs.

In 2022, Wynk Music migrated from a legacy autoscaling solution that was limiting Amazon EC2 Spot Instances adoption due to latency. It transitioned to Karpenter, a Kubernetes autoscaling solution that would simplify operations, maximize compute utilization, and reduce scheduling latencies. During the transition, Wynk Music personnel worked closely with AWS Solutions Architects and its Enterprise Support Technical Account Manager (TAM) to ensure best practices and to minimize risk. Comments Bhatia, “We deployed Karpenter successfully after our engineering team and AWS Enterprise Support worked together to fine-tune our architecture.”

Wynk Music also migrated multiple Amazon EKS workloads to Amazon EC2 Instances powered by AWS Graviton Processors. As a result, it consolidated the cluster behind its mission-critical Aerospike database by more than 66 percent—from 48 Intel virtual CPUs (vCPUs) to 12 Graviton vCPUs, reducing the instance size from m5.4xlarge to m6g.xlarge. In addition, the team saw latency for Aerospike reduced by around 20–25 percent[illegible]by 75 percent.

Over time, Wynk Music migrated 30 percent of its total workloads to Graviton-based Amazon EC2 Instances, reducing computing costs by 70 percent and lowering expenses for API services by 47 percent. “When one-third of our cost is allocated to Amazon EC2 instances, opting for AWS Graviton-based instances was a logical step due to its superior performance and cost-effectiveness,” says Bhatia.

In 2023, Wynk Music increased the performance of its big data analytics platform, which monitors user sessions for its 75 million monthly active customers. It replaced its Amazon EMR environment by transitioning from Amazon EC2 to Amazon EKS, utilizing Karpenter for provisioning and scaling. It also moved the platform to AWS Graviton, lowering analytics costs by 60 percent. Most recently, the company has migrated ScyllaDB, Kafka, ELK stack, Solr, and MongoDB workloads to AWS Graviton. “We’ve seen performance improvements across the board and made AWS Graviton part of our default setup for data service,” states Bhatia.

Furthermore, in collaboration with ADL, the Enterprise Support TAM implemented a strategic fortnightly cost-centric cadence, incorporating insights from Amazon EC2 Spot instances and AWS Graviton usage into a custom cost dashboard. These initiatives have resulted in substantial cost efficiencies, helping ADL effectively optimize the value of its technological investments.

Outcome | Accelerating Growth, Improving Services, and Driving Sustainability

Through optimization of its AWS infrastructure, Wynk Music has reduced costs per user by 20 percent, freeing up additional resources for continued development of its machine learning and data analytics capabilities, which recommend music to listeners. “By using AWS, we continue to increase the personalization of our recommendations to help retain customers and attract new ones,” says Bhatia.

Despite continued growth, the company remains aligned to its objective of sustainable IT. Through AWS Graviton, the business is consolidating clusters, reducing energy consumption, and increasing performance. “All business units across Airtel Digital take their environmental responsibilities seriously,” says Bhatia, “Using AWS, we can continuously raise the bar on sustainable innovation.”

About Wynk Music

Wynk Music, owned by Airtel Digital Ltd, is a streaming service for Indian music, with songs in 14 different languages. Launched in September 2014, Wynk Music is available across India, Sri Lanka, and 15 African countries.

AWS Services Used

Amazon EKS

Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run e AWS cloud and on-premises data centers.

Learn more »

Amazon EC2 Spot Instances

Amazon EC2 Spot Instances let you take advantage of unused EC2 capacity in the AWS cloud and are available at up to a 90% discount compared to On-Demand prices.

Learn more »

AWS Graviton Processors

AWS Graviton is a family of processors designed to deliver the best price performance for your cloud workloads running in Amazon Elastic Compute Cloud (Amazon EC2).

Learn more »

Amazon EMR on EKS

Amazon EMR on EKS provides a deployment option for Amazon EMR that allows you to run open-source big data frameworks on Amazon Elastic Kubernetes Service (Amazon EKS).

Learn more »

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